Progressive exercises

E1 · Direct application. Determine $\lim_{x \to +\infty} \dfrac{5x^3 - x}{2x^3 + 7}$ with justification.

E2 · Application and justification. Let f the function defined on $\mathbb{R} \setminus \{2\}$ by $f(x) = \frac{1}{(x-2)^2}$. Determine its limits at 2^-, at 2^+, at $-\infty$ and at $+\infty$.

E3 · Method to organise. Determine $\lim_{x \to 1} \dfrac{x^2 - 1}{x - 1}$ specifying why the simplification used is valid near 1.

E4 · Choice of strategy. Determine $\lim_{x \to +\infty} x^3 e^{-2x}$. State the comparative growth result used.

E5 · Synthesis or proof. Determine $\lim_{x \to -\infty} \left(\sqrt{x^2 + 1} + x \right)$. Justify in particular the treatment of $\sqrt{x^2}$.